THE MASTERMIND COACHING CLASSES
CBSE / RBSE • NURSERY – CLASS XII • JAIPUR
Road No. 5, VKIA, Jaipur — 9887258679

# ANSWER KEY & MARKING SCHEME SET 1

Chapter 7: **Measurement of Time and Motion** | Class 7 Science | Maximum Marks: 40

## SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(c)** 60 km/h | 90/1.5. |
| 2 | **(a)** 50 km/h | 100/2. |
| 3 | **(d)** speed | Rise over run. |
| 4 | **(c)** 9 km/h | 1.5 x 6. |
| 5 | **(b)** 2 h | 60/30. |
| 6 | **(a)** 5 m/s | 20/4. |
| 7 | **(c)** 10 m/s | 36 x 5/18. |
| 8 | **(c)** 8 m/s | 100/12.5. |
| 9 | **(d)** faster | 10 m/s versus 5 m/s. |
| 10 | **(a)** 90 km/h | 25 x 18/5. |

## SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | 150/2.5 = 60 km/h (1); 60 x 5/18 = 16.7 m/s (approx.) (1). | 1+1 |
| 12 | Effort and gravity act against uphill (1); Speed changes so motion is non-uniform (1). | 1+1 |
| 13 | Larger angles can change motion (1); Time period measurement is easier (1). | 1+1 |
| 14 | Time = 225/90 (1); 2.5 h (1). | 1+1 |

## SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | 40 min = 2/3 h (1); 12 / (2/3) = 18 km/h (1); 5 m/s (1). | 1+1+1 |
| 16 | Going 3/5 = 0.6 h (1); Returning 3/3 = 1 h (1); Total 1.6 h (1). | 1+1+1 |
| 17 | First 0.5 m/s (1); Second 0.75 m/s (1); Speed increased (1). | 1+1+1 |
| | **OR:** 200/20 = 10 m/s (1); 10 x 18/5 (1); 36 km/h (1). | 1+1+1 |
| 18 | Traffic (1); Signals (1); Stops and starts (1). | 1+1+1 |
| | **OR:** Accuracy (1); Avoid human delay (1); Fair results (1). | 1+1+1 |

## SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | Total time 4 h (1); Average 240/4 = 60 km/h (1); Running time 3.5 h (1); 240/3.5 = 68.6 km/h (1); Running speed is higher (1). | 1+1+1+1+1 |
| | **OR:** Friction and air resistance (1); Amplitude decreases (1); Spring or weights give energy (1); Regular winding (1); Clock continues (1). | 1+1+1+1+1 |
| 20 | 40 km in first 2 h (1); Horizontal for 1 h (1); 60 km more in 2 h (1); Total 100 km in 5 h (1); Average 20 km/h (1). | 1+1+1+1+1 |
| | **OR:** Cheetah 25 m/s (1); Car 20 m/s (1); Cheetah faster (1); Convert 25 m/s to 90 km/h (1); Compare (1). | 1+1+1+1+1 |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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## ANSWER KEY & MARKING SCHEME SET 2

Chapter 7: **Measurement of Time and Motion** | Class 7 Science | Maximum Marks: 40

### SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(c)** half | Time inversely proportional. |
| 2 | **(a)** the object at rest 30 m away | Distance fixed. |
| 3 | **(c)** 200 m | 20 m/s x 10 s. |
| 4 | **(b)** double | Proportional. |
| 5 | **(a)** speed increasing | Slope increases. |
| 6 | **(b)** 2.5 s | 60/24. |
| 7 | **(b)** 15 | 54 x 5/18. |
| 8 | **(c)** distance rising and then falling | Displacement returns. |
| 9 | **(c)** 40 oscillations | 60/1.5. |
| 10 | **(d)** 10 m/s | 100/10. |

### SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | Measure the distance and time to cover it (1); Speed = distance/time (1). | 1+1 |
| 12 | Multiply by 5/18 (1); 1000 m per 3600 s (1). | 1+1 |
| 13 | T = 50/25 = 2 s (1); 180/2 = 90 oscillations (1). | 1+1 |
| 14 | Speed keeps changing (1); Total distance divided by total time (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | Time 2 h + 1 h = 3 h (1); Distance 120 km (1); 40 km/h (1). | 1+1+1 |
| 16 | Container with a small hole (1); Mark water level (1); Time between marks (1). | 1+1+1 |
| 17 | Speedometer shows instantaneous speed (1); Average is total distance/time (1); They may differ (1). | 1+1+1 |
|  | **OR:** A: 50 km/h (1); B: 50 km/h (1); Equal speeds (1). | 1+1+1 |
| 18 | 72 km/h = 20 m/s (1); 36 km/h = 10 m/s (1); 10 < 15 < 20 (1). | 1+1+1 |
|  | **OR:** Measure a 100 m track (1); Time your walk (1); Divide (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | As L increases, T increases (1); Longer pendulum is slower (1); Mass kept same (1); Time 20 oscillations for accuracy (1); Conclude T depends on length only (1). | 1+1+1+1+1 |
|  | **OR:** Start gun triggers (1); Sensors at the finish (1); Time to 0.01 s (1); Human reaction slower (1); Accuracy (1). | 1+1+1+1+1 |
| 20 | Sundial: gnomon and shadow (1); Water clock: flow of water (1); Sundial works by day only (1); Water clock works day and night (1); Both less accurate (1). | 1+1+1+1+1 |
|  | **OR:** Two bobs of different masses (1); Same length (1); Count 20 oscillations each (1); Same time (1); Mass has no effect (1). | 1+1+1+1+1 |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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## ANSWER KEY & MARKING SCHEME SET 3

Chapter 7: **Measurement of Time and Motion** | Class 7 Science | Maximum Marks: 40

### SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(b)** non-uniform motion | Varying speed. |
| 2 | **(b)** the second hand of a clock | Constant speed. |
| 3 | **(c)** 1.5 h | 0.5 + 1. |
| 4 | **(c)** measuring 20 oscillations | Reduces error. |
| 5 | **(a)** 50 m | 5 m/s x 10. |
| 6 | **(d)** 90 km/h | 90 km/h = 25 m/s. |
| 7 | **(a)** increases | Longer swings slower. |
| 8 | **(a)** 2 km | 4 x 0.5. |
| 9 | **(a)** 3 m/s | 18 km/h = 5 m/s; 3 m/s smallest. |
| 10 | **(b)** remains same | Independent of mass. |

### SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | Total distance 20 m (1); 20/10 = 2 m/s (1). | 1+1 |
| 12 | 8 min = 2/15 h (1); 2 / (2/15) = 15 km/h (1). | 1+1 |
| 13 | Subtract the initial reading from the final (1); Difference is distance (1). | 1+1 |
| 14 | Quartz oscillates regularly (1); Sundial depends on sunlight (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | Time 20 oscillations (1); T = 36/20 (1); 1.8 s (1). | 1+1+1 |
| 16 | 16/10 = 1.6 s (1); 32/20 = 1.6 s (1); T = 1.6 s in both cases (1). | 1+1+1 |
| 17 | Equal swing times (1); Count swings (1); Clock use (1). | 1+1+1 |
| | **OR:** 8/2 = 4 m/s (1); 16/4 = 4 m/s (1); Uniform (1). | 1+1+1 |
| 18 | Repeats in equal times (1); Speed changes (1); Fast at the lowest point (1). | 1+1+1 |
| | **OR:** Needs clear shadow (1); No sunlight (1); Shadow unclear (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | Start: speed increasing (1); Straight track: uniform (1); Stop: speed decreasing (1); Whole journey non-uniform (1); Graph is curved at ends (1). | 1+1+1+1+1 |
| | **OR:** Time out 2 h (1); Time back 1 h (1); Total distance 120 km (1); 120/3 = 40 km/h (1); Not 45 (1). | 1+1+1+1+1 |
| 20 | Walk 900/600 = 1.5 m/s (1); Run 900/180 = 5 m/s (1); Total 1800 m in 780 s (1); 2.3 m/s (1); Between the two (1). | 1+1+1+1+1 |
| | **OR:** Plot points (1); Join (1); Slope gives speed (1); Flat part is rest (1); Compare sections (1). | 1+1+1+1+1 |

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